

Timertest:

```
✓ TimerTest (com 2 sec 258 ms) ✓ 11 tests passed 11 tests total, 2 sec 258 ms
  ✓ testPauseAndLap 275 ms
  ✓ testPauseAndLap 314 ms
  ✓ testLap 209 ms
  ✓ testPause 211 ms
  ✓ testStop 108 ms
  ✓ testMillisecs 105 ms
  ✓ testRepeat1 125 ms
  ✓ testRepeat2 247 ms
  ✓ testRepeat3 193 ms
  ✓ testRepeat4 361 ms
  ✓ testPauseAndLap 110 ms

/Users/Liang.zim/Library/Java/JavaVirtualMachines/ms-21.0.9/Contents/Home/bin/java ...

Process finished with exit code 0
```

InsertionsortTest

```
✓ InsertionSortTest (co 138 ms) ✓ 11 tests passed 11 tests total, 138 ms
  ✓ testMutatingInsert 111 ms
  ✓ sort0a 6 ms
  ✓ sort0b 0 ms
  ✓ sort0c 0 ms
  ✓ sort4a 1 ms
  ✓ sort4b 2 ms
  ✓ sort0 1 ms
  ✓ sort1 7 ms
  ✓ sort2 6 ms
  ✓ sort3 3 ms
  ✓ testStaticInsertionSc 1 ms

/Users/Liang.zim/Library/Java/JavaVirtualMachines/ms-21.0.9/Contents/Home/bin/java ...

Helper for InsertionSort with 4 elements
StatPack {hits: n=1; mean=7362; normalized=15.986; lookups: n=1; mean=2520; normalized=5.472; copies: n=1; mean=0; n
StatPack {hits: n=1; mean=14851; normalized=32.249; lookups: n=1; mean=4951; normalized=10.751; copies: n=1; mean=0;

Process finished with exit code 0
```

Benchmark result:

```
=== Array size: 100 ===
Random: 0.1910833 ms
Ordered: 0.0115333 ms
Partial: 0.0423834 ms
Reverse: 0.3588958 ms
```

```
=== Array size: 200 ===
Random: 0.968875 ms
Ordered: 0.0182292 ms
Partial: 0.2599959 ms
Reverse: 2.3330709 ms
```

```
=== Array size: 400 ===
Random: 0.3677334 ms
Ordered: 0.0187916 ms
Partial: 0.0812084 ms
Reverse: 0.4564584 ms
```

```
=== Array size: 800 ===
Random: 0.72465 ms
Ordered: 0.0085084 ms
Partial: 0.192225 ms
Reverse: 1.6191916 ms
```

```
=== Array size: 1600 ===  
Random: 3.0725917 ms  
Ordered: 0.0095833 ms  
Partial: 0.8823584 ms  
Reverse: 6.5181916 ms
```

Conclusion:

Insertion sort exhibits $O(n)$ performance for ordered arrays (time doubles when n doubles) and $O(n^2)$ performance for random and reverse-ordered arrays (time quadruples when n doubles).